

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
8	(a)		Speed of infra-red = $[(7.5 \times 10^{-7}) \times (4 \times 10^{14})] = 3 \times 10^8$ (1) Speed of microwaves = 3×10^8 (1) ecf Wavelength of microwaves = $\frac{3 \times 10^8 \text{ ecf}}{3 \times 10^{10}}$ (1) = 1×10^{-2} [m] (1) Award 1 mark for answer of 1×10^n where $n \neq -2$	1	1 1 1		4	4	
	(b)	(i)	Speed of infra-red waves in glass = $\frac{3 \times 10^8 \text{ ecf}}{1.5} = 2 \times 10^8$ m/s (1) ecf Calculation of the time delays as 0.24 [s] [by satellite] and 0.045 [s] [by cable] (1) Alternative: $\frac{72\,000}{9000} = 8$ (1) So the time delay is $\frac{8}{1.5} = 5.3$ times longer [by satellite] (1) accept the inverse argument		2		2	1	
		(ii)	Any 2 × (1) from: - Less time delay - More security - Very low error rate / less data loss - Greater data transfer rates (10 Gb/s) compared to satellites (50 Mb/s) / higher bandwidth - Less interference Accept less power loss	2			2		
		(iii)	Orbit above equator (1) [of Earth] Orbit period of 24 hour OR has to be 36 000 km above the Earth (1)	2			2		
			Question 8 total	5	5	0	10	5	0